Theory of Generated Space (TGP):

A minimal core of axioms, substrate, and effective field

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Abstract

We present the core of the *Theory of Generated Space* (TGP): a scalar-field theory in which space is not a background arena but is *constituted* by mass-energy sources through a shared field Φ. The theory is built on four axioms (absolute reference state N_0, $\mathbb{Z}_2$-symmetric relational substrate, matter as source, instability of N_0) and a single dynamical degree of freedom. From these inputs we derive: (i) a unique second-order kinetic operator with coefficient $\alpha = 2$, supplied directly by the Ginzburg–Landau substrate bond in (3) (continuum form in App. B, Prop. **??**) and confirmed by a classification theorem within conditions (C1)–(C3) of Sec. 4; (ii) an effective spatial metric of exponential form $g_{ij} = e^{+2U/c_0^2}\delta_{ij}$ that reproduces all post-Newtonian parameters of general relativity exactly ($\gamma_{\rm PPN} = \beta_{\rm PPN} = 1$); (iii) three interaction regimes (long-range attraction, intermediate repulsion, short-range confinement) from a single non-linear potential $U(\Phi)$; (iv) identically luminal gravitational-wave propagation $c_{\rm GW} = c_0$, consistent with GW170817. TGP is *not* a scalar-tensor theory: the metric is algebraically determined by Φ, and the action contains no Einstein–Hilbert term. This document defines the minimal, self-contained core; applications (superconductivity, lepton masses, black-hole shadows, dark energy) are developed in companion papers that cite this reference. We state twenty-one proven or numerically verified results and ten explicit open problems, so that the theory is falsifiable and extendable on a well-defined basis.

Keywords: generated space, emergent gravity, scalar field, $\mathbb{Z}_2$ substrate, PPN parameters, effective metric.

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1 Introduction

The standard model of physics, combined with ΛCDM cosmology and general relativity, is phenomenologically successful but treats space as a pre-existing manifold on which fields propagate. Approximately thirty-five free parameters govern the combined description, and the origin of the small cosmological constant, the hierarchy of lepton masses, and the mechanism of high-temperature superconductivity remain open.

We propose an alternative foundation in which space is not pre-given. Instead, every source of mass-energy *produces* a local contribution to a shared scalar field Φ, whose value measures the density of generated space. The metric, causality and Newtonian gravity emerge as effective descriptions of the configuration of Φ; the Planck length is the only invariant carried across scales.

The purpose of this paper is to *define* the theory — its axioms, substrate, effective field, and canonical equations — in a minimal, self-contained form, and to delineate explicitly what has been proven and what remains open. We consciously avoid a survey of applications: these are developed in companion papers that cite this reference (see Sec. 8).

What TGP is not. TGP is not a scalar-tensor theory: the effective metric is algebraically determined by Φ, and the action contains no Einstein–Hilbert term. TGP is not a programme in which space emerges from information or entanglement (AdS/CFT, ER = EPR): here space emerges from *matter*. TGP is not an extension of general relativity with additional fields; it is a reformulation in which GR is recovered in a well-defined regime.

What TGP is. A single scalar degree of freedom Φ, coarse-grained from a $\mathbb{Z}_2$-symmetric relational substrate, satisfying a single nonlinear field equation sourced by matter, and inducing an exponential effective metric. From this minimal structure flow all predictions of the theory.

Scope of this paper. Sections 2–5 establish the axiomatic base, the substrate-to-field derivation, the effective field theory, and the emergent metric. Section 6 exhibits the three physical regimes. Section 7 enumerates the status of theorems and open problems — the central artifact of this paper from a reproducibility standpoint. Section 8 is a brief forward reference to companion papers.

Main open caveat. The passage from the discrete substrate Hamiltonian H_Γ to the continuum TGP action rests on the Ginzburg–Landau (GL) bond in (3), whose continuum form $\mathcal{F}^{\rm geo}_{\rm kin}[\varphi] = (K_{\rm geo}/2)\int \varphi^4(\nabla\varphi)^2\, d^dx$ is an elementary Taylor expansion (App. B, Prop. **??**). What remains open is the strictly rigorous Banach/Γ-convergence statement for the block-spin map; this is a mathematical hardening task, not an open derivation of α. An earlier attempt to derive $\alpha = 2$ from a bilinear v1 bond (OP-6 of v1) was shown to fail on three independent grounds and has been resolved by the axiom pivot documented in Remark 10 and `KNOWN_ISSUES.md`. Throughout the paper we present the continuum field theory of Sec. 4 as a well-posed effective theory whose axiomatic continuum limit is explicit.

2 Axioms

TGP rests on four axioms. Axioms 1–5 concern the reference state and its instability; Axiom 4 states how matter couples to space. A fifth, derived statement (the $\mathbb{Z}_2$ symmetry of the substrate) appears as a proposition rather than a postulate.

Axiom 1 (Absolute reference state). *There exists a reference state N_0* – absolute nothingness – *characterized by the simultaneous vanishing of the state function $\mathcal{F}$ and the field Φ:*

$$N_0 := \{\emptyset,\ \mathcal{F}\}, \qquad \mathcal{F}\big|_{N_0} = 0, \qquad \Phi\big|_{N_0} = 0. \tag{1}$$

In N_0 there is no metric, no matter, no time, no energy. N_0 is not *the quantum vacuum and* not *empty space; it is the absence of every physical metric structure.*

Axiom 2 (Relational substrate). *Underlying both the N_0 state and all physical configurations is a fundamental, non-emergent relational substrate*

$$\Gamma = (V, E), \tag{2}$$

with vertices V and edges E, possessing adjacency relations and connectivity but no a priori metric. *Each node $i \in V$ carries an amplitude operator $\hat{s}_i$. The substrate Hamiltonian is*

$$H_\Gamma = \sum_i \Big[\tfrac{\hat{\pi}_i^2}{2\mu} + \tfrac{m_0^2}{2}\hat{s}_i^2 + \tfrac{\lambda_0}{4}\hat{s}_i^4\Big] + J\sum_{\langle ij\rangle} A_{ij}\,\hat{s}_i^2\,\hat{s}_j^2\,\big(\hat{s}_j^2 - \hat{s}_i^2\big)^2, \tag{3}$$

with $J > 0$, $\lambda_0 > 0$, and $A_{ij} \in \{0,1\}$ the adjacency matrix. The bond is a Ginzburg–Landau (GL) gradient term in the Z_2-even composite $\hat{\Phi}_i \equiv \hat{s}_i^2$, equivalent in continuum to $\mathcal{F}_{\text{kin}}^{\text{geo}}[\varphi] = (K_{\text{geo}}/2)\int \varphi^4(\nabla\varphi)^2\,d^dx$ (see App. B, Prop. **??***); it vanishes identically on any configuration with $\hat{s}_i = 0$, encoding the fact that* an empty substrate cannot support geometry. *This is a change from v1: see 10 below and* `KNOWN_ISSUES.md`.

Proposition 3 ($\mathbb{Z}_2$ symmetry). *The Hamiltonian* (3) *is invariant under the chiral inversion $\hat{s}_i \mapsto -\hat{s}_i$ for all $i \in V$. This discrete $\mathbb{Z}_2$ symmetry is the fundamental symmetry of TGP; the state function $\mathcal{F}$ measures the extent of its spontaneous breaking.*

Axiom 4 (Matter generates space). *Every source of mass-energy ρ_a contributes to the field Φ according to*

$$\mathcal{D}[\Phi_a] \;=\; q_a\,\rho_a, \tag{4}$$

where $q_a > 0$ is the spatial generation constant *of particle type a and $\mathcal{D}$ is the covariant operator governing the dynamics of Φ (made explicit in Sec. 4).*

Axiom 5 (Instability of N_0). *The state $\mathcal{F} = 0$ (N_0) is unstable: its measure in configuration space vanishes,*

$$\mu\big(\{\sigma \in \Omega : \mathcal{F}[\sigma] = 0\}\big) = 0. \tag{5}$$

Any departure from exact $\mathbb{Z}_2$ symmetry leads locally to $\mathcal{F} > 0$ and hence to simultaneous matter and generated space.

Proposition 6 (Instability from substrate thermodynamics). *Let $\mu_\beta(\sigma) \propto e^{-\beta H_\Gamma[\sigma]}$ be the Gibbs measure for* (3). *In the thermodynamic limit $|V| \to \infty$ there exists $\beta_c > 0$ such that $\mu_\beta\big(\{\mathcal{F} = 0\}\big) = 0$ for all $\beta > \beta_c$; the physical substrate ($\beta \to \infty$) is always in the spontaneously broken regime. Thus Axiom 5 is a* ***consequence*** *of Axioms 2–4 together with standard Ising thermodynamics, not an independent postulate.*

Remark 7 (Three-level ontology). *These axioms generate a three-level hierarchy: (0) the relational substrate Γ is fundamental and non-emergent; (1) the field Φ is a coarse-grained* metric-phase parameter *of the substrate, with $\Phi = 0$ the non-metric phase and $\Phi > 0$ the metric phase; (2) the effective geometry $g_{\mu\nu}$ arises from coarse-graining level (1). The transition $\Phi = 0 \to \Phi > 0$ does not create space from nothing: it switches on the metric response of an already existing substrate, in the same sense that freezing water does not create molecules but changes their phase.*

3 From substrate to field

The relation between the microscopic substrate Γ and the macroscopic field Φ is established by coarse-graining. Given a block $\mathcal{B}(\mathbf{x})$ of lattice sites centred on $\mathbf{x}$,

$$\Phi(\mathbf{x}) \;\propto\; \langle \hat{s}^2 \rangle_{\mathcal{B}(\mathbf{x})}. \tag{6}$$

In the mean-field approximation one has the identity

$$\mathcal{F}_i[\Omega] \;=\; \langle \Omega | \, \hat{s}_i \, | \Omega \rangle^2, \tag{7}$$

so that $\mathcal{F} = 0$ iff $\langle \hat{s}_i \rangle = 0$ (unbroken $\mathbb{Z}_2$, i.e. N_0), while $\mathcal{F} > 0$ signals spontaneous breaking and the simultaneous appearance of matter and space. In the continuum limit, the bond term in (3) reduces to the Ginzburg–Landau gradient functional $\mathcal{F}^{\rm geo}_{\rm kin}[\varphi] = (K_{\rm geo}/2) \int \varphi^4 (\nabla\varphi)^2 d^d x$, as made explicit in App. B (Prop. **??**); the Euler–Lagrange equation of this functional is the continuum field equation for Φ that we state and analyse in Sec. 4.

Remark 8 (One substrate, two projections)**.** *Coarse-graining H_Γ yields* two *emergent fields from a single microscopic object:*

(i) the scalar metric-phase parameter $\Phi(\mathbf{x}) \propto \langle \hat{s}^2 \rangle_{\mathcal{B}}$, which governs the entire gravitational/cosmological sector;

(ii) a spatial-correlation tensor $\sigma_{ab}(\mathbf{x}) \propto \langle \hat{s}_i \, \hat{s}_{i+\hat{e}_a} \rangle^{\rm TF}_{\mathcal{B}}$, carrying spin-2 and underlying tensor gravitational-wave modes.

Both follow from the same *H_Γ and are not independent postulates. This is crucial: TGP is not a scalar-tensor theory, because Φ and σ_{ab} are two projections of one substrate rather than two independent fields.*

4 Effective field theory

We use the dimensionless variable $\varphi \equiv \Phi/\Phi_0$.

4.1 Uniqueness of the kinetic operator

Theorem 9 (Uniqueness of $\mathcal{D}_{\rm kin}$ within class (C1)–(C3), $\alpha = 2$)**.** *Among all local second-order scalar kinetic operators of the form*

$$\mathcal{D}[\varphi] = K(\varphi)\, \nabla^2 \varphi + \tfrac{1}{2} K'(\varphi) (\nabla\varphi)^2, \tag{8}$$

imposing

(C1) $K(\varphi)$ smooth and depending only on φ (locality);

(C2) $\mathcal{D}[\varphi] \to \nabla^2\varphi$ for $\varphi = \text{const}$ (Laplacian limit);

(C3) covariance under conformal rescaling $\varphi \mapsto \lambda\varphi$ of the substrate metric $h_{ij} = \varphi^4 \, \delta_{ij}$ (Φ-covariance),

selects uniquely $K(\varphi) = K_{\rm geo}\varphi^4$ with the ratio $\alpha(\varphi) \equiv K'(\varphi)\varphi/[2K(\varphi)] = 2$ (constant). The kinetic operator is

$$\boxed{\mathcal{D}_{\rm kin}[\varphi] \;=\; \nabla^2\varphi + \frac{2(\nabla\varphi)^2}{\varphi} \;=\; \frac{1}{3\varphi^2}\, \nabla^2(\varphi^3).} \tag{9}$$

Proof sketch. Condition (C2) fixes $K(1) = 1$. The Euler–Lagrange equation of $\mathcal{L}_{\rm kin} = -\frac{1}{2} K(\varphi)(\nabla\varphi)^2$ yields the ratio $\alpha(\varphi) = K'(\varphi)\varphi/[2K(\varphi)]$. (C1) forces α to be a constant; (C3) fixes $\alpha = 2$ through the explicit computation of the Laplace–Beltrami operator on $h_{ij} = \varphi^4 \, \delta_{ij}$: $\Delta_h \varphi = \varphi^{-4} \, \mathcal{D}_{\rm kin}[\varphi]$ holds iff $\alpha = 2$. Hence $K = \varphi^4$ (up to the overall constant $K_{\rm geo}$, conventionally 1). □

Remark 10 (What $\alpha = 2$ means; closure of OP-6)**.** *The exponent $\alpha = 2$ is* not *fitted to data: it is fixed by the geometric coupling $K_{ij} = J(\varphi_i\varphi_j)^2$ between nearest neighbours on the substrate, under the three conditions (C1)–(C3). The statement is therefore a uniqueness result* within *the stated class of local, Φ-covariant second-order scalar operators; relaxing any of (C1)–(C3) (non-locality, higher derivatives, or departures from the conformal rescaling of the substrate metric) would open additional channels that are not considered by TGP.*

In v1 of this paper (Zenodo DOI `10.5281/zenodo.19670324`*) the substrate Hamiltonian carried a linear bond $-J\hat{s}_i\hat{s}_j$; the statement that the continuum limit generates the GL coupling $K_{ij} = J(\varphi_i\varphi_j)^2$, and thus supplies (C1)–(C3) from below, was listed as open problem OP-6. Between 2026-04-23 and 2026-04-24, OP-6 was investigated on three independent grounds: (a) the analytical sketch of standard RG mechanisms (M2-a); (b) 1D Monte Carlo of the envelope stiffness and of the block-RG bond flow (M2-b, M3-a); (c) scaling-dimension analysis at the 3D Ising Wilson–Fisher fixed point using conformal-bootstrap data (M3-c). All three rule out the v1 derivation: at 3D Ising WF, the ratio $K^{(1)}/K^{(0)}$ decays in the IR as $k^{\Delta_\epsilon} = k^{1.41}$, so $K_*(\varphi) \to$ const rather than $K_*(\varphi) \propto \varphi$. Numerically, the 1D block-RG measurement returns $K_B(\langle\Phi_B\rangle) \propto \langle\Phi_B\rangle^{-0.52\pm0.28}$ uniformly across block sizes $B \in \{1, 2, 4, 8, 16\}$, rejecting the target $p_\Phi = +1$ at 5.5σ.*

The present paper (v2 in preparation) therefore adopts the GL bond directly as the substrate axiom (3)*. Under this axiom, (C1)–(C3) and the GL gradient functional $\mathcal{F}^{\rm geo}_{\rm kin}$ of Prop.* **??** *(App. B) are* immediate*, and Theorem 9 reduces to the statement that any axiomatic choice respecting substrate–metric conformal covariance must be of this GL form. OP-6 is closed (see* `TGP_v1/research/op6/` *for the complete record). The bond vanishes on empty-substrate configurations ($\hat{s}_i = 0$), making "no substrate $\Rightarrow$ no geometry" a property of the axiom rather than a derived fact.*

4.2 Action, potential, and field equation

The complete TGP action is

$$S_{\rm TGP}[\Phi, \psi_m] = \int d^4x\, \sqrt{-g_{\rm eff}}\, \Big[\mathcal{L}_{\rm field}(\Phi) + \mathcal{L}_{\rm mat}(\Phi, \psi_m)\Big], \tag{10}$$

with field Lagrangian

$$\mathcal{L}_{\rm field} = -\tfrac{1}{2}\, K(\varphi)\, g^{\mu\nu}_{\rm eff}\partial_\mu\varphi\, \partial_\nu\varphi - U(\varphi), \qquad K(\varphi) = \varphi^4, \tag{11}$$

self-interference potential

$$U(\varphi) = \frac{\beta}{3}\varphi^3 - \frac{\gamma}{4}\varphi^4, \qquad \beta = \gamma \ \ (\text{vacuum condition } U'(1) = 0), \tag{12}$$

and minimal matter coupling $\mathcal{L}_{\rm mat} = -\,q\,\varphi\rho/\Phi_0$. The volume element is $\sqrt{-g_{\rm eff}} = c_0\varphi$ (see Sec. 5).

Theorem 11 (Full TGP field equation)**.** *Variation $\delta S_{\rm TGP}/\delta\Phi = 0$ of* (10) *yields*

$$\boxed{\nabla^2\Phi + \frac{2(\nabla\Phi)^2}{\Phi} + \beta\,\frac{\Phi^2}{\Phi_0} - \gamma\,\frac{\Phi^3}{\Phi_0^2} \;=\; -\,q\,\Phi_0\,\rho.} \tag{13}$$

Remark 12 (TGP is not a scalar-tensor theory)**.** *The action* (10) *contains* no *Einstein–Hilbert term R. The effective metric $g^{\rm eff}_{\mu\nu}$ is defined algebraically by Φ (not an independent dynamical variable). The Einstein equations arise as a geometric consistency condition of $g^{\rm eff}_{\mu\nu}$, propagated by the Bianchi identity once satisfied at an initial time; they are not postulated. The only dynamical degree of freedom is Φ.*

Remark 13 (Formal contrast with scalar-tensor theories)**.** *Structurally TGP differs from the Brans–Dicke / Horndeski family on three counts, each visible at the level of the action and the equations of motion:*

(i) Independent tensor d.o.f. absent. *In scalar-tensor theories the action is* $S = \int \sqrt{-g}\,[f(\phi)R/16\pi G + \ldots]$ *with* $g_{\mu\nu}$ *varied independently. Here the only field varied is* Φ*; the metric is algebraically determined by* $g^{\rm eff}_{\mu\nu}[\Phi]$ *through Eq.* (17).

(ii) No conformal frame ambiguity. *Scalar-tensor theories admit Jordan/Einstein frame reparametrisations. TGP has a single frame: the substrate metric* $h_{ij} = \Phi^4\delta_{ij}$ *is fixed by the* $\mathbb{Z}_2$ *substrate structure; no conformal redefinition of the "physical" metric is available.*

(iii) Kinetic operator not a free function. *In scalar-tensor theories* $f(\phi)$ *is an input function to be fitted. Here* $K(\varphi) = \varphi^4$ *is fixed by Theorem 9 under (C1)–(C3), not chosen by hand.*

These three properties preclude TGP from being recast as a standard scalar-tensor theory in any frame; the GR limit is recovered only as an effective low-curvature description, not as a conformal transformation of a two-metric theory.

4.3 Ghost-freedom

The substrate coupling $K_{\rm sub}(\varphi) = \varphi^4$ is strictly positive for all $\varphi > 0$, so the kinetic term $\frac{1}{2}\varphi^4(\nabla\varphi)^2 \geq 0$ is positive-semidefinite. TGP is ghost-free at three levels simultaneously:

Sector	**Kinetic positivity**	**Mechanism**
Fundamental (φ)	$\varphi^4(\nabla\varphi)^2 > 0$	geometric
Mukhanov–Sasaki perturbations	$Q_s = \varphi^4_{\rm bg} > 0$	linearized
Solitons (full lattice)	$K_{\rm sub} = \varphi^4 > 0$	substrate

A logarithmic truncation $K = 1 + 4\ln\varphi$ that appears in certain expansions exhibits an apparent zero at $\varphi^* = e^{-1/4}$; this is an artifact of the Taylor expansion of the exponent, $\varphi^4 = e^{4\ln\varphi} = 1 + 4\ln\varphi + \ldots$, not a physical singularity.

5 Emergent metric

The effective metric of TGP is not postulated. It is derived from three inputs: (i) node density of the substrate $\Rightarrow g_{ij} = \psi\,\delta_{ij}$ with $\psi \equiv \Phi/\Phi_0$; (ii) information budget $\Rightarrow f \cdot h = 1$ (antipodal condition); (iii) PPN consistency $\Rightarrow$ exponential form.

5.1 Spatial part: node-density ansatz

A coarse-graining cell $\mathcal{B}(\mathbf{x})$ containing N_B lattice nodes has physical volume proportional to $\Phi(\mathbf{x})$:

$$g_{ij} = \frac{\Phi}{\Phi_0}\,\delta_{ij} = \psi\,\delta_{ij}, \qquad \psi \equiv \Phi/\Phi_0. \tag{14}$$

5.2 Temporal part: information budget

The substrate has a fixed information capacity $\mathcal{B} = N_B s_0$ per cell. Partitioning between spatial capacity $h(\Phi)$ and temporal capacity $f(\Phi)$, the multiplicative conservation law

$$f(\Phi) \cdot h(\Phi) = 1 \tag{15}$$

gives $f = \Phi_0/\Phi = 1/\psi$.

5.3 Exponential metric from PPN consistency

Theorem 14 (Selection of $h(\Phi) = \Phi$ and exponential form, within the TGP ansatz class)**.** *Within the ansatz class*

$$g_{tt} = -c_0^2/\psi, \qquad g_{ij} = h(\Phi)\,\delta_{ij}, \qquad fh = 1 \text{ (antipodal projection)}, \tag{16}$$

and in the weak-field limit $\Phi = \Phi_0(1 + 2U/c_0^2)$ with U the Newtonian potential, the power-law choice $h = (\Phi/\Phi_0)^p$ forces $\beta_{\rm PPN} = 2p$; PPN consistency $\beta_{\rm PPN} = 1$ then selects $p = 1$ uniquely. Continuing the same power-law structure to all *orders in U requires the exponential closure*

$$\boxed{ds^2 = -c_0^2\, e^{-2U/c_0^2}\, dt^2 + e^{+2U/c_0^2}\, \delta_{ij}\, dx^i dx^j, \qquad \psi = e^{+2U/c_0^2}.} \tag{17}$$

Both forms $f = 1/\psi$ and $f = e^{-2U/c_0^2}$ satisfy $fh = 1$ and agree at $O(U)$; the exponential form extends the agreement to all orders.

Remark 15 (Scope of the selection statement)**.** *Theorem 14 is a selection statement* within *the TGP ansatz class $fh = 1$ and power-law $h(\Phi)$. It is not a uniqueness statement against every conceivable (f, h); a parametrically distinct metric structure (e.g. bi-metric or tensor-valued) would not be in competition here. Five independent arguments support the choice $p = 1$ (substrate-density interpretation, volume element, Cassini/LLR PPN, soliton mass ratio, information budget); see* `research/metric_ansatz/` *for details.*

Corollary 16 (PPN parameters exact, within the ansatz of Theorem 14)**.** *As an algebraic consequence of the exponential ansatz* (17) *(selected within the TGP ansatz class; see Remark 15), expanding the metric and comparing with the standard PPN expansion, one obtains*

$$\gamma_{\rm PPN} = 1, \qquad \beta_{\rm PPN} = 1 \qquad \text{(exact to all orders in } U\text{)}. \tag{18}$$

All ten PPN parameters equal their GR values. The prediction is consistent with every current solar-system test: Cassini Shapiro-delay tracking gives $|\gamma_{\rm PPN} - 1| \le 2.3 \times 10^{-5}$ [7], and Lunar Laser Ranging (Nordtvedt effect) gives $|\beta_{\rm PPN} - 1| \le 6 \times 10^{-5}$ [8]; a dedicated cross-check on all ten PPN parameters is reported in `research/metric_ansatz/` *(8/8 PASS).*

Corollary 17 (Luminal gravitational waves)**.** *Gravitational waves propagate on the effective metric $g_{\rm eff}$ at the vacuum speed: $c_{\rm GW} = c_0$ identically. This is consistent with GW170817 constraints ($|c_{\rm GW} - c_0|/c_0 < 10^{-15}$).*

5.4 Field-dependent constants

Requiring invariance of the Planck length $\ell_{\rm P} = \sqrt{G\hbar/c^3}$ under the emergent rescaling fixes the field dependence of the fundamental constants:

$$c(\Phi) = \frac{c_0}{\sqrt{\psi}}, \qquad G(\Phi) = \frac{G_0}{\psi}, \qquad \hbar(\Phi) = \frac{\hbar_0}{\sqrt{\psi}}, \tag{19}$$

all consistent with the exponential metric (17). $\ell_{\rm P}$ is the sole invariant across scales.

6 Three physical regimes

The non-linear potential $U(\varphi)$ together with the source term in Eq. (13) generates three distinct interaction regimes from a single field equation. Linearization around a single static point source yields a radial profile $\phi(r)$ with the following properties:

Regime	Scale	Physical content
(i) long-range attraction	$r \gg r_{\rm rep}$	Newtonian gravity, $U \sim -GM/r$
(ii) intermediate repulsion	$r \sim r_{\rm rep}$	positive spatial potential between sources
(iii) short-range well	$r \lesssim r_{\rm well}$	confinement ($U \to -\infty$ finite well)

The existence of three regimes follows from the non-monotonicity of $\phi(r)$: a single generation profile acquires at least one inflection point from the interplay of $\mathcal{D}_{\rm kin}[\varphi]$ with the cubic and quartic terms of $U(\varphi)$. The explicit scales $r_{\rm rep}$ and $r_{\rm well}$ depend on the source strength qM/Φ_0 and on the coupling γ through

$$\Lambda_E \equiv \gamma/12, \qquad 3H_0^2 \approx c_0^2\,\Lambda_E \quad \text{(de Sitter limit)}, \tag{20}$$

which identifies Λ_E as the effective cosmological constant.

Remark 18 (Gravity as regime (i))**.** *Newtonian gravity is recovered as the long-range limit of the full non-linear equation* (13)*: in the weak field $\Phi \approx \Phi_0(1 + 2U/c_0^2)$, the leading source term reproduces Poisson's equation $\nabla^2 U = 4\pi G\rho$ with $G = q\Phi_0 c_0^2/(8\pi)$ (identifying q, Φ_0 with Newton's constant).*

7 Status of theorems and open problems

We distinguish three levels of status, as follows: [TH] formally proven theorem; [NU] numerically verified to stated precision; [HY] conjecture or hypothesis, not yet proven. The list below summarises the current (April 2026) status of TGP core.

7.1 Proven results (Core)

#	Result	Content	Status
1	Uniqueness of $\mathcal{D}_{\rm kin}$	Theorem 9: $K(\varphi) = \varphi^4$, $\alpha = 2$ unique under (C1)–(C3)	TH
2	Ghost-freedom (three sectors)	Sec. 4.3: $K_{\rm sub} > 0\ \forall\varphi > 0$; no kinetic singularity	TH
3	Full field equation	Theorem 11: Eq. (13) from $\delta S/\delta\Phi = 0$	TH
4	Exponential metric uniqueness	Theorem 14: PPN consistency $\Rightarrow$ exponential form	TH
5	PPN parameters exact	Corollary 16: $\gamma_{\rm PPN} = \beta_{\rm PPN} = 1$ to all orders	TH
6	Luminal gravitational waves	Corollary 17: $c_{\rm GW} = c_0$ identically (GW170817)	TH
7	Invariance of $\ell_{\rm P}$	Eq. (19): $\ell_{\rm P} = \sqrt{G\hbar/c^3} = $ const consistent with $g^{\rm eff}_{\mu\nu}$	TH
8	Instability of N_0	Proposition 6: from Ising thermodynamics of H_Γ	TH
9	$\mathbb{Z}_2$ substrate symmetry	Proposition 3: H_Γ invariant under $\hat{s}_i \mapsto -\hat{s}_i$	TH
10	Three regimes from $U(\varphi)$	Sec. 6: non-monotonic $\phi(r)$ from cubic+quartic U	TH
11	Bianchi propagation of Friedmann constraint	Remark following Eq. (13): $\nabla_\mu G^{\mu\nu}_{\rm eff} = 0$	TH
12	Dark-energy de Sitter limit	Eq. (20): $3H_0^2 \approx c_0^2\Lambda_E$, $\Lambda_E = \gamma/12$	TH
13	Matter-source covariance	Axiom 4 implemented by $\mathcal{L}_{\rm mat} = -q\varphi\rho/\Phi_0$	TH
14	Vacuum condition $\beta = \gamma$	From $U'(1) = 0$; Eq. (12)	TH
15	MS kinetic positivity in FRW	$Q_s = \varphi_{\rm bg}^4 > 0$ for all cosmological backgrounds	TH

7.2 Numerically verified (to stated precision)

#	Result	Content	Status
16	FRG structural stability of $K(\rho) \propto \rho$	One-loop LPA' flow near the Wilson–Fisher fixed point in $d = 3$, Z_2 class: the kinetic form $K(\rho) \propto \rho$ is preserved, $K_{\mathrm{IR}}/K_{\mathrm{UV}} = 1.000$, $\eta^* = 0.044$. This verifies kinetic stability of the axiomatic GL form of (3) at the continuum level; $\alpha = 2$ itself is supplied by the axiom (see Remark 10).	NU
17	Cosmology: $\Delta\chi^2 = +1.8$, $\Delta\mathrm{AIC} = +3.8$	vs. ΛCDM on Planck + DESI data, Friedmann (13)	NU
18	Solitonic N-body stability	Chaos suppression, Earnshaw violation; 10/10 tests pass	NU
19	Metric derivation chain	Spatial $\to$ information $\to$ PPN; cross-checked with $\ell_{\mathrm{P}} = \mathrm{const}$	NU
20	Three-regimes existence	Numerical profile $\phi(r)$ with attraction, repulsion, confinement	NU
21	Coarse-graining $H_\Gamma \to S_{\mathrm{TGP}}$ (axiomatic + numerical)	In the v2 axiom, the GL bond of (3) is by construction the lattice version of $\mathcal{F}^{\mathrm{geo}}_{\mathrm{kin}}[\varphi] = (K_{\mathrm{geo}}/2)\int \varphi^4(\nabla\varphi)^2 d^dx$ (App. B, Prop. **??**); no numerical extraction of α is required. Numerical block-spin Monte Carlo studies that originally probed the v1 bilinear bond (`research/continuum_limit/cg_strong_numerical.py`, `TGP_v1/research/op6/m3a_block_rg_1d.py`) are retained as supporting evidence for the axiom-level pivot; the complete record is in `KNOWN_ISSUES.md` (2026-04-24) and in `TGP_v1/research/op6/`.	NU

7.3 Open problems

The following are explicitly open: they define the research frontier of TGP at the core level.

#	Open problem	Statement	Status
OP-1	Uniqueness of $U(\varphi)$	Is the cubic+quartic form the unique self-interference potential compatible with $\mathbb{Z}_2$ symmetry and the vacuum condition?	HY
OP-2	Parametric reduction $\beta = \gamma$	Formal proof (beyond the vacuum condition) that $\beta = \gamma$ is protected at all scales.	HY
OP-3	Identification $a_\Gamma = 1/\Phi_0$	Whether the two fitted parameters coincide, reducing the free input count to one.	HY
OP-4	Derivation of the quartic coefficient γ	An *a priori* determination of γ (equivalently Λ_E) from H_Γ alone.	HY
OP-5	Full non-linear N-body regime	Existence/uniqueness of solutions for strongly overlapping sources beyond perturbation theory.	HY

#	Open problem	Statement	Statu
OP-6	Substrate metric continuum limit (rigorous)	Rigorous Banach contraction of the block-spin map and Γ-convergence of the energy functional from the v2 GL bond (3) to the continuum action (10). **Axiomatic closure: done** (2026-04-24, axiom pivot; see Remark 10 and `KNOWN_ISSUES.md`). Mathematical hardening (strict Banach/Γ-convergence theorem): remains open.	HY
OP-7	Tensor projection σ_{ab}	Derivation of the explicit tensor-field equation from the same H_Γ; GW polarisation content.	HY
OP-8	Gauge-sector emergence	Emergence of U(1)×SU(2)×SU(3) from the substrate (addressed partially in the extended manuscript).	HY
OP-9	Strong-field black-hole solutions	Complete catalogue of non-singular BH solutions; quasi-normal modes.	HY
OP-10	Quantum theory of Φ	Second quantisation of Φ beyond the MS linear regime.	HY

8 Applications and falsifiability

The minimal core of TGP defined above motivates several companion studies in which the same substrate and exponential metric are confronted with separate empirical domains. Each companion paper cites this reference for its foundational content; the items below are pointers, not self-contained derivations.

- **Superconductivity.** Using the substrate harmonic $a^*_{\mathrm{TGP}} = 7.725\, a_{\mathrm{B}} \approx 4.088\,\text{Å}$ together with the orbital-amplitude spectrum $\{A_s, A_{sp}, A_d, A_f\}$, the companion paper [2] presents a single-formula fit of T_c across cuprates, phonon-mediated metals, pressure-driven systems and lanthanide hydrides in terms of three universal TGP constants plus tabulated atomic inputs. Scope, systematics and excluded families are discussed there.
- **Lepton masses.** A single substrate parameter $g_0^e = 0.86941$ reproduces m_μ/m_e and m_τ/m_e to better than 10^{-3}, and the Koide combination $K = 2/3$ is recovered as an arithmetic identity of the Brannen parametrisation [3]; the identification of g_0^e with a substrate quantity is a working hypothesis of that companion paper.
- **Black-hole shadows.** In the strong-field regime the exponential metric (17) leads to a photon-ring radius reduced by $\sim 37\%$ relative to Schwarzschild, offering a concrete target for next-generation EHT measurements [4].
- **N-body dynamics.** Numerical integration of Eq. (13) in configurations dominated by the intermediate-repulsion regime (regime ii) exhibits chaos suppression and a violation of Earnshaw's theorem; see [5] for the test suite.

Minimal falsification targets. The following predictions are of a binary character and provide the most direct falsification tests of the core theory:

(F1) $\alpha = 2$ *identically* (Theorem 9): any observable inconsistent with $K \propto \varphi^4$ at the substrate level falsifies TGP.

(F2) $\gamma_{\mathrm{PPN}} = \beta_{\mathrm{PPN}} = 1$ *to all orders* (Corollary 16): a measured deviation at $O(U^2/c_0^4)$ beyond GR would falsify the exponential metric.

(F3) $c_{\mathrm{GW}} = c_0$ *identically* (Corollary 17): already supported by GW170817 at the 10^{-15} level.

Closed-branch negative checks. Three potential near-term falsification channels have been analysed in dedicated companion studies and found to lie *below* current experimental precision, closing them as viable discovery avenues without contradicting the theory:

- *Muon* $(g-2)$*:* the TGP-substrate correction to the anomalous magnetic moment is $\Delta a_\mu \lesssim 10^{-12}$, three orders of magnitude below the current experimental uncertainty; see `research/muon_g_minus_`
- *Wiedemann–Franz in bad metals:* the substrate-diffusion correction to L/L_0 is $\mathcal{O}(10^{-4})$ at 300 K, three to four orders of magnitude smaller than observed deviations in cuprates and heavy-fermion compounds, which are therefore *not* of TGP origin; see `research/thermal_transport_mole`
- *Casimir force in MOFs:* the TGP correction is at the 10^{-4}–10^{-3} level, below current torsion-balance precision; see `research/casimir_mof/`.

These null results are stated not as corroborations but as explicit constraints on where a TGP-specific signature could plausibly emerge with present instrumentation.

9 Conclusion

We have presented a minimal, self-contained core of the Theory of Generated Space: four axioms, one scalar field with a geometrically fixed kinetic operator ($\alpha = 2$), a single non-linear potential with two coefficients related by a vacuum condition, and an algebraically determined effective metric reproducing general relativity exactly at the post-Newtonian level. Fifteen core results are formally proven, six more are numerically verified, and ten explicit open problems are stated so that the theory is both well-defined and falsifiable on a clear basis.

This core is intended to serve as a common reference for applications to superconductivity, particle masses, black-hole phenomenology and cosmology, developed in companion papers.

Data and reproducibility. All numerical scripts underlying the results of this paper are available in the accompanying source repository [1]. The present paper deposits a fixed, versioned snapshot on Zenodo (DOI assigned at publication).

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